C. Pranav Sharma

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Education

Amrita Vishwa Vidyapeetham Coimbatore, India
B.Tech in Computer Science and Engineering, CGPA: 8.13/10 *Aug 2023 – Apr 2027*

Relevant Coursework: Machine Learning, Probability and Statistics, Distributed Systems, Data Structures and Algorithms, Operating Systems, DBMS, Cyber Security

Technical Skills

Languages: Python, SQL, Java, C, JavaScript
Machine Learning: PyTorch, Hugging Face Transformers, XGBoost, scikit-learn, Optuna, NumPy, Pandas
MLOps and Deployment: MLflow, DVC, Docker, FastAPI, Streamlit, Model Registry, CI/CD
Data and Cloud: PostgreSQL, TimescaleDB, PostGIS, MongoDB, AWS EC2, S3, Cloudflare R2
Core CS: Data Structures and Algorithms, Operating Systems, DBMS, Statistics, OOP

Projects

GNSS Observatory: Large-Scale Anomaly Detection Platform May 2026 – Present
Python, TimescaleDB, PostGIS, FastAPI, Docker, AWS EC2, Cloudflare R2 gnssobservatory.com

- Independently designed and built a two-region geospatial platform that ingested **692M aircraft position fixes** and **173M navigation-integrity samples** across a **full year** of telemetry from 23,600 aircraft, reaching **98.1% temporal coverage** with zero unexplained gaps.
- Engineered the storage and feature layer in **TimescaleDB and PostGIS** using **H3 hexagonal spatial indexing**, aggregating raw telemetry into **52.6M cell-windows** with rolling 30-day statistical baselines as detector inputs.
- Developed **three unsupervised anomaly detectors** and a cross-signal fusion stage that surfaced **52 corroborated candidate events**, each written with a **SHA-256 hashed evidence bundle** to object storage for full reproducibility.
- Measured detector performance against **held-out control airspace**, reaching **70% detection sensitivity at a 1.5% false-positive rate**; an earlier corroboration rule requiring three or more distinct aircraft types cut false positives from **14.7% to 0.07%** where threshold tuning alone had plateaued at 4.8%.
- Provisioned and operated the full **AWS EC2** backfill environment with a **Dockerized pipeline** and versioned SQL migrations; identified **13 silent data-scoping defects** and added SQL-inspecting regression tests that assert query correctness rather than output values.

AegisBERT: Zero-Day Threat Intelligence Engine April 2026
PyTorch, Hugging Face Transformers, FastAPI, Docker, Streamlit github.com/c-pranav-sharma/AegisBERT

- Fine-tuned **ModernBERT** on **150,000+ custom-mined CVEs** across **13 threat categories**, automating vulnerability classification over unstructured security advisories.
- Engineered an **automated NVD ingestion pipeline** with hybrid weak labeling (CWE taxonomy mapping and regex heuristics), eliminating manual annotation across the entire training corpus.
- Optimized inference using **Flash Attention and mean pooling**, achieving **under 200 ms latency** at an **8192-token context window**.
- Shipped a **Dockerized microservice architecture** pairing an asynchronous FastAPI inference API with a Streamlit analytics dashboard for real-time triage.

SentinelScore: Production MLOps Credit Risk Engine March 2026
XGBoost, Optuna, MLflow, DVC, FastAPI, Docker github.com/c-pranav-sharma/credit-risk-engine

- Built an end-to-end credit risk scoring system over **300K+ loan records** with **120+ engineered features**, improving **PR-AUC by 18%** through Bayesian hyperparameter optimization with Optuna.
- Designed a **fully reproducible DVC pipeline** with stage-level dependency tracking, cutting redundant recomputation by roughly **60%** across experiment iterations.
- Integrated **MLflow experiment tracking and a model registry**, logging **50+ runs** with metrics, parameters, and feature-importance artifacts for auditable model selection.
- Deployed a **Dockerized FastAPI inference service** with dynamic feature reconstruction and **F1-optimized decision thresholds**, serving predictions in **under 120 ms**.

Achievements

- **Research paper accepted at IEEE I-SMAC 2026** — *"A Multi-Protocol Cyber-Deception Framework Combining Language-Model Response Synthesis with Blockchain-Anchored Forensic Logging"*, accepted for presentation and publication at the **10th International Conference on IoT in Social, Mobile, Analytics and Cloud (I-SMAC)**, Tribhuvan University, Nepal, September 2026.